AI Engineer Roadmap (Beginner to Job-Ready) – 2026

SUMMARY

This roadmap is designed for beginners and outlines a practical learning path from programming fundamentals to production-ready AI systems. Estimated Duration: 10–15 Months (2–4 hours/day).

INTRODUCTION

Artificial Intelligence Engineering combines software engineering, machine learning, deep learning, cloud computing, and Generative AI to build intelligent applications. This roadmap is designed for beginners and outlines a practical learning path from programming fundamentals to production-ready AI systems.

Estimated Duration: 10–15 Months (2–4 hours/day)

PHASE 1: PROGRAMMING & COMPUTER SCIENCE FUNDAMENTALS (1–2 MONTHS)

Before learning AI, build a strong programming foundation.

Learn: - Python Programming - Object-Oriented Programming (OOP) - Git & GitHub - Linux Command Line - Basic Computer Science Fundamentals - Basic Data Structures & Algorithms

Mini Projects: - Calculator - To-Do App - Student Management System

PHASE 2: MATHEMATICS & SQL (1 MONTH)

AI relies heavily on mathematics and data manipulation.

Mathematics: - Linear Algebra - Probability - Statistics - Basic Calculus - Gradient Descent

SQL: - SELECT, WHERE, GROUP BY - Joins - Aggregate Functions - Window Functions - Database Design Basics

PHASE 3: DATA ANALYSIS (3 WEEKS)

Learn to clean, analyze, and visualize data before training models.

Tools: - NumPy - Pandas - Matplotlib - Seaborn

Learn: - Data Cleaning - Exploratory Data Analysis (EDA) - Feature Engineering - Data Visualization

Projects: - Titanic Dataset Analysis - Sales Data Dashboard

PHASE 4: MACHINE LEARNING (2 MONTHS)

Master classical machine learning algorithms and model evaluation.

Learn: - Regression - Classification - Clustering - Feature Engineering - Model Evaluation - Hyperparameter Tuning

Library: - Scikit-learn

Projects: - House Price Prediction - Spam Detection - Customer Churn Prediction

PHASE 5: DEEP LEARNING (1–2 MONTHS)

Learn neural networks for solving complex AI problems.

Learn: - Neural Networks - CNNs - RNNs - Transformers - Transfer Learning

Frameworks: - PyTorch - TensorFlow

Projects: - Image Classification - Digit Recognition

PHASE 6: COMPUTER VISION & NLP (1–2 MONTHS)

Develop AI applications that understand images and language.

Computer Vision: - OpenCV - YOLO - Object Detection - Image Segmentation

Natural Language Processing: - Text Preprocessing - Hugging Face Transformers - Text Classification - Named Entity Recognition

Projects: - Face Mask Detection - Sentiment Analysis - News Classifier

PHASE 7: GENERATIVE AI (1–2 MONTHS)

Learn the technologies powering modern AI applications.

Learn: - Large Language Models (LLMs) - Prompt Engineering - Embeddings - Vector Databases - Retrieval-Augmented Generation (RAG) - AI Agents - Multi-Agent Systems

Frameworks: - LangChain - LangGraph - LlamaIndex

Vector Databases: - Astra DB - Pinecone - Qdrant - ChromaDB

Projects: - AI Chatbot - PDF Question Answering - AI Interview Assistant

PHASE 8: MLOPS & CLOUD (1 MONTH)

Learn to deploy and manage AI applications in production.

Learn: - MLflow - Docker - FastAPI - CI/CD Basics - Cloud Deployment

Cloud Platforms: - AWS - Microsoft Azure - Google Cloud Platform

Project: - Deploy an AI application using Docker and MLflow.

PHASE 9: BUILD YOUR PORTFOLIO

Create projects that demonstrate practical AI engineering skills.

Recommended Portfolio Projects: - Medical Image Classification - AI Resume Screening System - RAG-based Knowledge Assistant - AI Interview Preparation Platform - Multi-Agent AI System - AI Document Generation Application

Upload all projects to GitHub, write detailed documentation, and deploy at least two applications.

PHASE 10: INTERVIEW PREPARATION

Revise technical concepts and practice solving interview questions.

Focus Areas: - Python - SQL - Data Structures & Algorithms - Machine Learning - Deep Learning - Computer Vision - NLP - LLMs - RAG - LangChain - LangGraph - MLOps - System Design - Behavioral Questions

Practice coding regularly, explain your projects confidently, and participate in mock interviews.

FINAL ADVICE

Focus on understanding concepts rather than memorizing them. Build projects after completing each major topic, maintain an active GitHub portfolio, stay updated with the latest AI advancements, and continuously improve your software engineering skills. Employers value practical experience, problem-solving ability, and well-documented projects as much as theoretical knowledge. Following this roadmap consistently will prepare you for entry-level AI Engineer roles in 2026.